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XLRS-Bench: Could Your Multimodal LLMs Understand Extremely Large Ultra-High-Resolution Remote Sensing Imagery?

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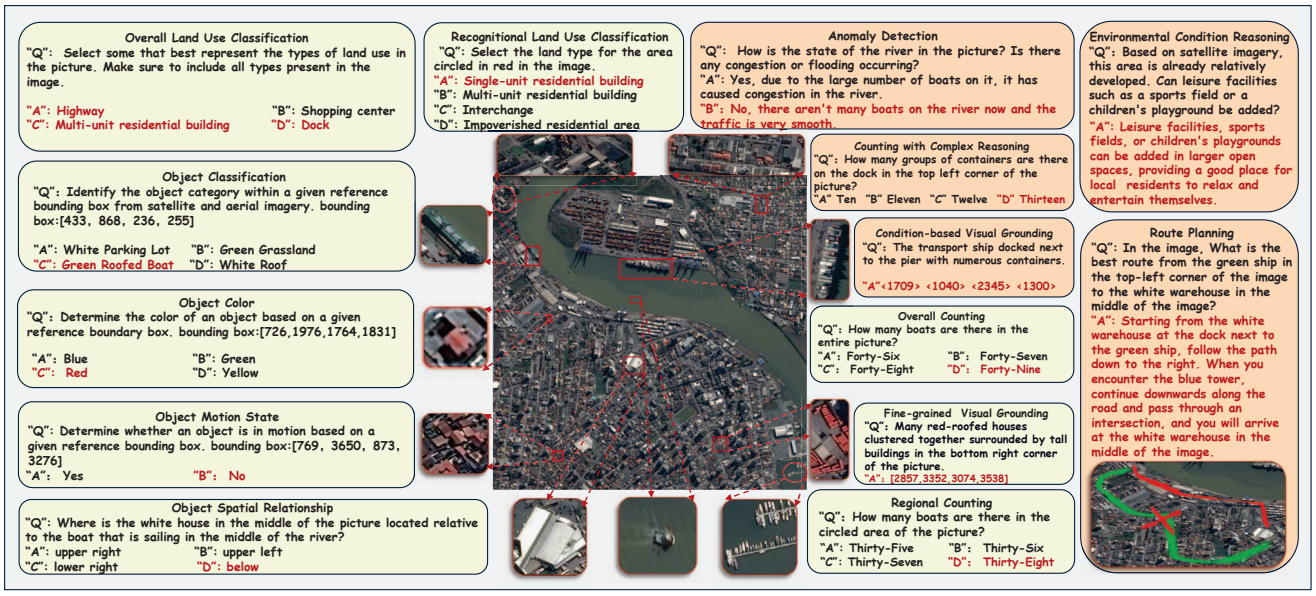

Figure 1. A typical example from our XLRS-Bench. XLRS-Bench focuses on extremely large ultra-high-resolution RS imagery, integrating over 10 multimodal vision-language perception and reasoning tasks within the same image.

Abstract

The astonishing breakthrough of multimodal large language models (MLLMs) has necessitated new benchmarks to quantitatively assess their capabilities, reveal their limitations, and indicate future research directions. However, this is challenging in the context of remote sensing (RS), since the imagery features ultra-high resolution that incorporates extremely complex semantic relationships. Existing benchmarks usually adopt notably smaller image sizes than real-world RS scenarios, suffer from limited annotation quality, and consider insufficient dimensions of evaluation.

To address these issues, we present XLRS-Bench: a comprehensive benchmark for evaluating the perception and reasoning capabilities of MLLMs in ultra-high-resolution RS scenarios. XLRS-Bench boasts the largest average image size (8500×8500) observed thus far, with all evaluation samples meticulously annotated manually, assisted by a novel semi-automatic captioner on ultra-high-resolution RS images. On top of the XLRS-Bench, 16 sub-tasks are defined to evaluate MLLMs' 10 kinds of perceptual capabilities and 6 kinds of reasoning capabilities, with a primary emphasis on advanced cognitive processes that facilitate real-world decision-making and the capture of spatiotemporal changes. The results of both general and RS-focused MLLMs on XLRS-Bench indicate that further ef-

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forts are needed for real-world RS applications. We have open-sourced XLRS-Bench to support further research in developing more powerful MLLMs for remote sensing.

1. Introduction

Recent advancements in multimodal large language models (MLLMs) [2, 41, 51, 54, 75] have significantly enhanced visual understanding and reasoning. As real-world applications require more detailed visual processing, many MLLMs [15, 28, 31, 67, 71] have been developed to improve understanding of high-resolution images. To fully assess and leverage their potential, benchmarking is essential, leading to the creation of various related datasets [5, 12, 13, 24, 33, 52, 62, 72].

Remote sensing (RS) images have become essential for monitoring and understanding human environments, driving advancements in applications like precision agriculture [70], urban planning [76], and disaster assessment [11]. As such, assessing the performance of MLLMs in this field is crucial. However, the high resolution and complex semantic relationships in RS imagery make evaluating MLLMs in real-world remote sensing contexts particularly challenging. While recent studies [27, 38, 69] have proposed benchmarks and metrics to assess MLLM performance in RS, these efforts remain limited in three key areas:

Image Size. Real ultra-high resolution RS images often capture scenes at the least city-level or above with large size (*e.g.*, 10,000×10,000). However, benchmarks like VRS-Bench [27] merely utilize 512×512 image slices to evaluate the performance of MLLMs on cross-modal perception and understanding (like visual question answering (VQA), image captioning and visual grounding), which fails to provide a comprehensive assessment of the models' capabilities to capture long-range spatial-semantic relationships.

Human Annotation. The prohibitive labor costs associated with comprehensive manual annotation significantly limit the scalability of existing RS multimodal benchmarks, such as LHRS-Bench [38] (only 108 RS images paired with 690 QA pairs). In response, GPT-based annotators have become widely employed to automate data construction for both training and evaluating MLLMs, thus generating substantial data volumes. Following this trend, larger-scale RS benchmarks have been created [27]. However, the absence of stringent manual calibration in automated annotation processes may lead to hallucinations and errors, compromising the quality of the data. Additionally, empirical evidence [72] indicates that benchmarks annotated by MLLMs, such as GPT-4V, often incorporate linguistic biases, which may inadvertently boost the performance of corresponding models on these benchmarks, despite attempting to manually adjust content biases.

Evaluation Dimension. Existing RS benchmarks [27, 38, 69] primarily concentrate on basic perceptual capabilities such as attribute recognition, semantic discrimination, and spatial localization. However, existing benchmarks fail to fully assess MLLMs' interactive perception and complex reasoning, which are crucial for tasks like visual grounding, path planning, and intent inference.

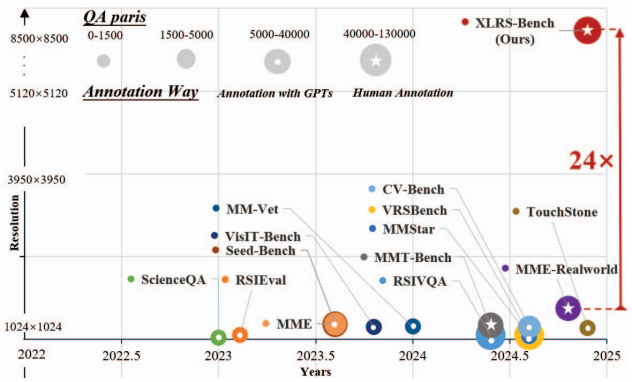

Figure 2. Advantages of XLRS-Bench: XLRS-Bench boasts an average image size that is 24 times larger than existing datasets.

To address these challenges, we introduce XLRS-Bench, a benchmark designed to assess the perception and reasoning capabilities of MLLMs on ultra-high-resolution RS scenarios. We first collected 1,400 real-world ultra-high-resolution RS images with large sizes (8,500×8,500 pixels on average). Among them, 530 images come from detection datasets (*e.g.*, DOTA-v2 [60]) and 870 images are from segmentation datasets (*e.g.*, MiniFrance [6]). Then, 45 experts are involved in hand-crafted annotation and cross-verification for 16 sub-tasks. The annotations are presented as three vision-language formats including VQA, captioning and visual grounding. Specifically, we implemented a semi-automated pipeline for image captioning, by leveraging GPT-4o for pre-annotation and human for verification. We additionally invite 10 specialists in the field of MLLM for further quality control. Finally, XLRS-Bench incorporates 45,942 annotations covering 10 ability indicators for perception and 6 for reasoning, comprising 32,389 VQA pairs, 12,619 visual grounding instances, and 934 detailed captions, as shown in Fig. 1. Based on XLRS-Bench, we evaluated a series of general MLLMs and an RS-specific MLLM and conducted an in-depth analysis. As shown in Fig. 2, the key advantages of XLRS-Bench over existing benchmarks include:

- **Ultra-high Resolution**. XLRS-Bench features the largest image sizes available, 10∼20× than that of existing datasets, with 840 images out of all images at a resolution of 10,000×10,000 pixels.
- **High-quality Annotation**. All the annotations are human involved and manually verified through iterations, resulting in a high-quality benchmark for evaluating MLLMs on real ultra-high-resolution RS scenarios.

- **Comprehensive Evaluation Dimensions**: XLRS-Bench covers 10 perception indicators and 6 reasoning dimensions to assess MLLMs' capabilities, encompassing 16 sub-tasks with a total of 45,942 questions. Especially, XLRS-Bench includes complex reasoning tasks to explore MLLMs' potential in conducting planning and change detection in long spatial-temporal RS scenarios.

In summary, our main contributions are as follows:

- We establish XLRS-Bench, a new benchmark for assessing the perception and reasoning capabilities of MLLMs across 16 sub-tasks in ultra-high-resolution RS scenarios.
- We develop a semi-automated pipeline to expand detailed captions, showcasing the potential of the scalability across various RS tasks.
- We evaluate prominent MLLMs across computer vision and RS fields on XLRS-Bench, providing insights into future directions for understanding real RS scenarios.

2. Related Work

General Multimodal Benchmark. Large-scale vision-language models (VLMs) have demonstrated great potential in multimodal tasks like complex scene understanding and visual sentiment analysis. Recently, multimodal evaluation datasets have been developed to assess the capabilities of these models quantitatively. However, previous benchmarks mainly focused on specific domains and assessed models on one or a few tasks (*e.g.*, captioning [1, 8, 63] or VQA [14, 16, 21, 37, 48, 74]). The evaluation for large-scale VLMs requires more comprehensive benchmarks. Therefore, MME [12] offers a benchmark spanning 14 perceptual and cognitive tasks, and MMBench [33] includes over 3,000 questions across 20 skill dimensions, like object localization and social reasoning. Seed-Bench [24] expands the sample size with 19,000 questions, and MMT-Bench [62] incorporates real-world data from areas like autonomous driving and embedded AI. MME-Realworld [72], extending to five real-world scenarios, provides the highest resolution natural scene benchmark so far, with images at 2,000×1,500. However, these general-purpose benchmarks have two main limitations: limited data and text annotations for RS scenarios, and smaller image sizes compared to real RS data. Even MME-Realworld, with the most RS data, includes only three types of QA pairs that are related to RS scenarios. Furthermore, its average resolution of 2,000×1,500 is still far below the required of real RS tasks (*e.g.*, HRSCD [10] at 10,000×10,000 for RS segmentation).

Remote Sensing Multimodal Benchmark. With the recent advancements of large multimodal models in general domains, the RS field has also experienced rapid growth on MLLMs [22, 38]. This has led to the creations of relevant evaluation benchmarks. RSIEval [19] offers 100 human-annotated captions and 936 visual question-answer pairs, mainly for image captioning and VQA tasks. LHRS-Bench [38] provides 108 images and 690 questions, with VQA questions spanning five dimensions. What's more, RSSA [42] introduces a benchmark focused on hallucination, while FIT-RSRC [36] targets understanding object relationships in RS scenes. VRSBench [27] includes 29,614 images with paired descriptions, 52,472 object references, and 123,221 question-answer pairs. Nevertheless, these benchmarks have two main limitations. First, they lack a diverse range of sub-tasks, resulting in less objective and comprehensive evaluations. Second, their image sizes are typically limited to 512×512, which is far smaller than the requirements of real RS tasks (*e.g.*, DOTA [60] images reach 7,000×4,000 for detection tasks, HRSCD [10] images reach 10,000×10,000 for segmentation), hindering the evaluation of MLLMs for long-range spatial-semantic cognition in real-world RS scenarios.

Multimodal Large Language Model. Leveraging advanced LLMs like GPTs [41] and LLaMA [54], MLLMs have shown impressive capabilities [55]. Proprietary models such as Gemini [51] and GPT-4o [41] exhibit strong understanding and reasoning skills, while open-source models, including Qwen-VL [2], InternLM-XComposer [67], MiniCPM [18], LLaVA [30] and MiniGPT-4 [75], also demonstrate notable performance. Since not specifically optimized for high-resolution tasks, these models only support image resolutions from 2K to 4K. Recently, several MLLMs have addressed low-resolution constraints to enable high-resolution processing. For example, LLaVA-Next [31] divides high-resolution images into patches, encodes each patch independently, and then links the patch tokens with the global image tokens. Furthermore, models such as Monkey [29] and LLaVA-UHD [15] compress these patches to avoid redundant tokens. Alternatively, Mini-Gemini [28] uses a dual encoder—one for high-resolution images and another for low-resolution embeddings, while Cambrian [53] uses learnable latent queries to interact with multiple visual features, and SliME [71] compresses local patches twice, preserving global features. In the RS field [20, 23, 56, 57], MLLMs have advanced as well. Geochat [22], based on LLaVA-1.5 [54], enables multi-task dialogues with RS images. LHRS-Bot [38] enhances image understanding through a multi-level vision-language alignment strategy and curriculum learning, and EarthGPT [69] unifies multisensor interpretation tasks for general RS understanding.

Despite these advancements, both RS and general-domain MLLMs face challenges. General MLLMs, while focused on high-resolution processing, lack rigorous testing on larger-size benchmarks in real-world RS scenarios. Meanwhile, RS models have only been tested on benchmarks with small images (*e.g.*, 512×512). In contrast, our XLRS-Bench provides a rigorous test bed to assess models' perception and reasoning capabilities on large-size images

from real RS environments.

3. XLRS-Bench

XLRS-Bench stands out from existing multimodal understanding benchmarks with four key features: i) the largest average image size of 8,500×8,500, featuring extensive 10,000×10,000 imagery; ii) 16 sub-tasks designed to evaluate MLLMs' capabilities in ultra-high-resolution RS scenes; iii) strict quality control to ensure sample accuracy and validity; iv) bilingual support for evaluating VLM performance in both English and Chinese. Further details on the construction of XLRS-Bench are provided below.

3.1. Evaluation Dimensions

Humans have exceptional perception and reasoning abilities that are foundational to complex cognition. Perception gathers sensory input, while reasoning draws conclusions. Together, these abilities enable tasks like object recognition, problem-solving, and decision-making. To achieve true artificial general intelligence in the real ultra large-scale RS scenarios, MLLMs must also exhibit robust perception and reasoning capabilities. We therefore classify perception and reasoning as primary (**L-1**) abilities, further refined into 11 **L-2** and 16 **L-3** capability dimensions. Figure 3 illustrates this classification, whose detailed explanations are presented in the following text.

3.1.1. Perception

We employ two typical tasks: image captioning and VQA, to measure the L-1 perception ability. To assess the capabilities in capturing fine-grained information for large-size ultra-high-resolution RS scenes, at the L-2 level, we additionally definite 6 evaluation aspects, which are further divided into 10 more refined indicators in L-3 grade. Next, we separately introduce the L-2 capabilities, with the L-3 tasks are also presented.

- **Image Captioning.** In this task, we go beyond simple single-sentence captioning tasks to *Detailed Image Captioning*, as large-size ultra-high-resolution RS images contain abundant details.
- **Scene Classification.** Scene Classification is evaluated through two L-3 sub-tasks: *Overall Land Use Classification* and *Regional Land Use Classification*. The former focuses on global image information with multiple-choice questions, while the latter examines detailed scene understanding in localized areas using single-choice questions.
- **Counting.** This capability contains two L-3 dimensions: *Overall Counting and Regional Counting*. Overall Counting requires comprehensive object recognition for the whole image, which is challenging even for humans. Regional Counting narrows the focus to smaller areas.
- **Object Spatial Relationship.** This dimension requires the model to identify two specified objects and determine their spatial relationship, demanding advanced spatial perception and understanding ability.

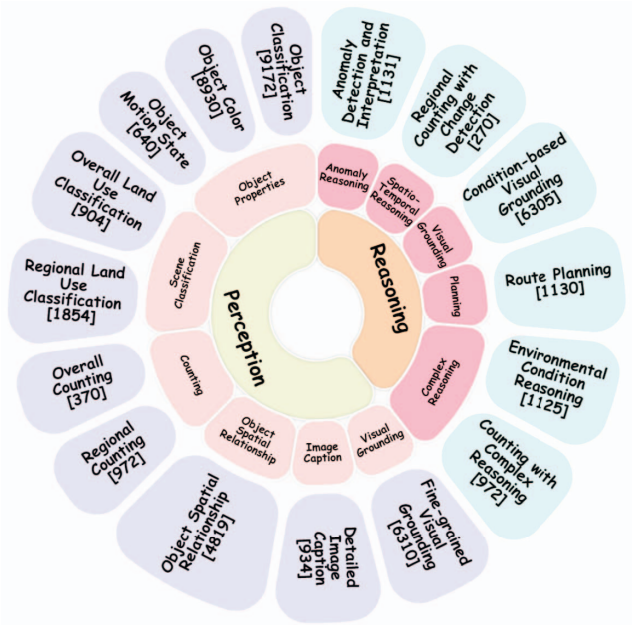

Figure 3. XLRS-Bench evaluates the perception and reasoning capabilities of MLLMs across three levels and 16 sub-tasks.

- **Object Properties.** Object understanding is assessed across three L-3 dimensions: *Object Classification, Object Color, and Object Motion State.* We exclude properties like shape and size [27], as they fall under Object Classification. Here, we conduct a separate evaluation of Object Color, since it differs significantly from attributes like shape and size. Object Motion State is especially relevant, as it tests reasoning through context clues, such as inferring a ship's movement from its wake.
- **Visual Grounding.** This dimension tests the model's ability to precisely locate objects. We assess *Fine-grained Visual Grounding*, which is essential in RS, where the model is required to detect small object types in large-scale high-resolution scenarios. Notably, some objects in these large-size images may be as small as 5-10 pixels, which presents a challenge for current MLLMs that primarily support 2K resolution, compressing such objects to just 1-2 pixels.

3.1.2. Reasoning

Unlike previous assessments of simple reasoning [27], we would like to evaluate the model's ability to perform Complex Reasoning based on rich visual information in ultra-high-resolution RS scenarios. We designed challenging 5 L-2 dimensions and 6 L-3 dimensions to test reasoning in realistic ultra-high-resolution RS scenarios:

- **Complex Reasoning.** It is divided into two L-3 dimensions, *Environmental Condition Reasoning* and *Counting with Complex Reasoning*. The former dimension

Table 1. Comparison between existing vision-language benchmarks and our benchmark. ✅, ⊗ and ✗separately represent the annotations are machine generated, manually written and semi-automated, *i.e.*, machine generation followed by human verification.

Dataset	Average Resolution	Remote Sensing	Detailed Caption		Visual Grounding		VQA	
			Volume	Average Length	Volume	Annotation Method	Volume	Annotation Method
MMStar [7]	512×375	N	-	-	-	⊗	1,500	⊗
ScienceQA [47]	378×249	N	-	-	-	⊗	21,000	⊗
MM-Vet [64]	1,200×675	N	-	-	-	⊗	218	⊗
Seed-Bench [24]	1,024×931	N	-	-	-	⊗	19,242	⊗
MMT-Bench [62]	2,365×377	N	-	-	-	⊗	32,325	⊗
TouchStone [3]	897×803	N	-	-	-	⊗	908	⊗
VisIT-Bench [5]	765×1,024	N	-	-	-	⊗	1,159	⊗
BLINK [13]	620×1,024	N	-	-	-	⊗	3,807	⊗
CV-Bench [52]	1,024×768	N	-	-	-	⊗	2,638	⊗
MME [12]	1,161×840	N	-	-	-	⊗	2,374	✅
MMBench [33]	512×270	N	-	-	-	⊗	3,217	✅
MME-Realworld [72]	2,000×1,500	N	-	-	-	⊗	29,429	✅
UCM-Captions [44]	250×250	Y	10,500	12	-	⊗	-	⊗
RSICD [35]	500×500	Y	54,605	12	-	⊗	-	⊗
RSICap [19]	512×512	Y	2,585	60	-	⊗	-	⊗
RSVG [49]	800×800	Y	-	-	7,933	✗	-	⊗
DIOR-RSVG [65]	800×800	Y	-	-	38,320	✗	-	⊗
RRSIS-D [32]	800×800	Y	-	-	17,402	✗	-	⊗
RSVQA-HR [34]	1,024×1,024	Y	-	-	-	⊗	1,066,316	⊗
RSIVQA [73]	512×512	Y	-	-	-	⊗	111,134	✗
RSIEval [19]	512×512	Y	-	-	-	⊗	933	✗
VRSBench [27]	512×512	Y	29,614	52	52,472	✗	123,231	✗
XLRS-Bench	8,500 ×8,500	Y	934	379	12,619	✅	32,389	✅

leverages the comprehension information in ultra-high-resolution images, requiring the model to make contextual inferences. *Counting with Complex Reasoning* requires not only perception-based counting but also the ability to perform complex reasoning.

- **Anomaly Reasoning.** The model is expected to detect anomalies and predict potential risks using contextual clues, named *Anomaly Detection and Interpretation*. For instance, in a town near a mountain, the model should identify exposed forest land and, considering climate conditions, predict possible landslide risks.
- **Planning.** Beyond reasoning, the model should plan a route based on provided conditions, named *Route Planning* (L-3 ability). Given a starting point described in natural language, the model must locate it accurately on the image and select an optimal route based on criteria like shortest distance or minimal intersections.
- **Visual Grounding.** We also include conditional grounding to evaluate reasoning-based localization abilities, named *Condition-based Visual Grounding* (L-3 ability). Here, the model locates objects by inferring based on complex conditions in the image.
- **Spatiotemporal Reasoning.** The *Regional Counting with Change Detection* is unique in the RS field, as it evaluates changes in the number of objects between two temporal images of the same location.

3.2. Data Collection and Quality Control

Data Source To create a benchmark with diverse evaluation dimensions, we collected large-size high-resolution RS images rich in visual details, in order to design varied and challenging assessment tasks. Specifically, we compiled 1,400 images from realistic RS scenarios for different downstream tasks, selecting them rigorously based on diversity and quality. For detection tasks, we sourced 270 images at 4,096×4,096 and 210 images at 7,360×4,912 from DOTA-v2 [60], and added 50 images at a size of 3,744×5,616 from the ITCVD [61] dataset. For segmentation tasks, we used 457 images at 10,000×10,000 resolution from MiniFrance [6], 13 images at 11,500×7,500 from Toronto [46], and 30 images at 6,000×6,000 from Potsdam [45]. Additionally, for change detection tasks, we included 185 pairs (370 images) at 10,000×10,000 resolution from the HRSCD [10] dataset. Each image underwent multi-round cross-validation by annotators to ensure a rich, diverse selection of large-size and ultra-high-resolution samples. To be noticed, our dataset includes 840 images at 10,000×10,000 pixels—a significant milestone in benchmark works. To the best of our knowledge, this is the first large-scale evaluation using 10,000×10,000 images. Although challenging for human evaluation, we believe it is an essential step toward realistic RS applications.

Visual Question Answering In XLRS-Bench, we compile visual language question-answer pairs in a multiple-choice format for each L-3 capability, excluding grounding and captioning dimensions. Each pair $(P_i = (Q_i, C_i, I_i, T_i))$ includes: Q_i (the question), C_i (a set of n options, $2 \leq n \leq 4$), I_i (the relevant image), and T_i (the correct answer). It is important to note that research indicates GPT-based tools can introduce bias in benchmarks, favoring similar models and compromising their rigor and real-world assessment capacity [72]. To address this, trained annotators expand

the question set through human-only annotation, avoiding the assistance of GPTs. To minimize design bias, we implemented cross-validation and formed an external review team. Specifically, three annotation groups, each with 15 members, cross-validated each other's work, while a review team of 10 experts with MLLM experience resolved discrepancies and validated question design, ensuring that all questions genuinely assess MLLM capabilities.

Visual Grounding In XLRS-Bench, we evaluated visual grounding across two L-3 dimensions. We analyze each image by selecting 5-10 objects per L-3 dimension, resulting in 10-20 objects per image. Concretely, we carefully crafted reference sentences for each object to enable independent and precise identification. Distinctive features in these sentences such as color, shape, position, size, and relative spatial attributes distinguished each reference object from others. For the *Condition-based Visual Grounding* task, annotators used detailed image information to create specific descriptions of conditions, including possible constraints on state or geographic details. Notably, all visual grounding annotations were performed by the annotation and quality control team of the VQA group, ensuring high-quality manual annotation throughout.

Image Caption Unlike other approaches that assess MLLM's understanding of RS images with single, simple descriptions, we believe the rich details in ultra-high-resolution images require thorough and precise descriptions. For captioning, we use a semi-automated pipeline to interpret and generate text annotations of the image, as shown in Figure 4. We begin by dividing each image into nine sub-images, as using GPT-based tools to annotate the entire image often results in low-quality captions that lose important details, whereas the sub-image approach helps mitigate this issue.

In the next steps, we input 10 images (nine sub-images plus a compressed full image) into GPT-4o using complex prompts to generate detailed and comprehensive captions. For instance, for a 10,000 $\times$ 10,000 image where a river spans the scene with a wetland on one side and urban areas on the other, using sub-images alone cannot reflect the river's flow and overall layout, while only adopting the full image lacks local details. Our combined input method addresses both problems. For the ultra-high-resolution RS scenarios, we adopt a general-to-specific structure that separately describes the overall image and each sub-image, where the key objects in the description of sub-images (like vehicles, boats, and buildings) are counted. Then we require the GPT-4o to count and display prominent objects across the entire image, identify anomalies, and predict potential risks or developments. Detailed captions and multi-image inputs resulted in extensive token usage with GPT-4o, costing over $2,000 for captioning 1,000 images. However, despite precise prompts, GPT-4 still struggles with complex

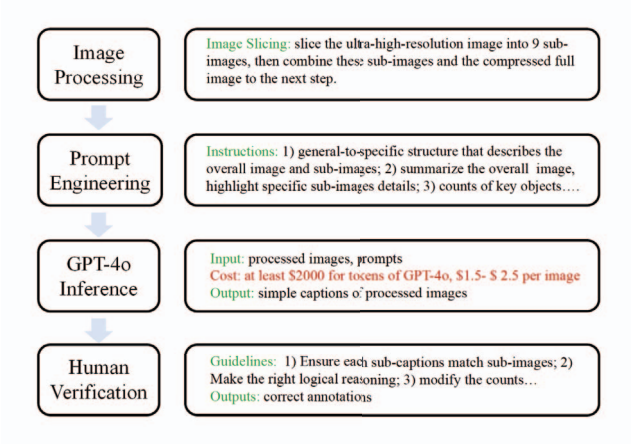

Figure 4. Semi-automated pipeline for detailed image captioning in XLRS-Bench.

tasks such as counting and reasoning, necessitating substantial human intervention to ensure the quality of the captions. To address this, annotators refine the captions using existing VQA annotations for detail and quality control. These refinements include adjusting summaries, correcting object counts, refining scene types, ordering regional descriptions by prominence, and manually annotating anomalies.

3.3. Analysis

Resolution. There are some benchmarks that adopt the image with sizes up to 2,000 $\times$ 1,500 pixels, but they are primarily designed for natural images rather than RS scenes. In contrast, current RS benchmarks typically remain below 512 $\times$ 512 pixels, as shown in Table 1. In contrast, our XLRS-Bench, with an average size of 8,500 $\times$ 8,500 pixels, provides extensive views that facilitate a deeper understanding of real RS scenarios. Remarkably, it includes 840 images at 10,000$\times$ 10,000 pixels, producing over 10,000 QA pairs. In fact, even annotators must carefully analyze these images to answer questions, with cross-validation employed to reduce errors. We hope this large-size RS benchmark is able to drive the development of MLLMs.

Length of Caption. XLRS-Bench offers comprehensive descriptions that cover global image context, regional details, specific object attributes, object counts, anomalies, and reasoning information. Each caption starts with a general image overview, followed by precise details across various regions. Regional details include object attributes such as count, color, shape, size, and spatial positioning—both absolute values within the image and relative to other objects like buildings, roads, and trees. In summary, these descriptions emphasize clear, distinct features, avoiding ambiguity, and typically average 19 sentences (379 words) in English and 20 sentences (663 words) in Chinese.

Bilingual. Traditional VQA methods [33] use translation engines to convert QA pairs from English to Chinese,

Table 2. Experimental results on the perception and reasoning dimensions of VQA tasks, with models ranked by average performance. Proprietary models are highlighted in gray. 'Avg' represents the average accuracy across sub-tasks.

Method	LLM	Language	Perception				Reasoning				
Sub-tasks (L-2 Capability)			Counting	Scene Classification	Object Spatial Relationship	Object Properties	Complex Reasoning	Planning	Spatiotemporal Reasoning	Anomaly Reasoning	Avg.
CogVLM2	Llama3-8B	en	36.66	46.26	**35.92**	**36.89**	60.23	34.16	-	69.85	**39.80**
Qwen2-VL	Qwen2-7B	en	**39.72**	48.30	32.04	35.07	64.23	32.12	**45.93**	68.35	38.99
LLaVA-OneVision	Qwen2-7B	en	35.84	47.76	32.60	35.95	60.37	24.07	37.78	71.88	38.90
GPT-4o-mini	-	en	28.32	45.46	29.96	36.29	54.84	33.81	15.19	72.06	37.99
InternVL2	InternLM2.5-7B	en	33.91	44.93	26.60	34.84	**66.43**	33.01	44.44	**74.54**	37.89
InternLM-XComposer-2.5	InternLM2-7B	en	35.84	47.43	32.75	30.60	64.76	35.31	32.22	69.50	36.33
LLaVA-Next	Llama3-8B	en	38.00	40.79	32.50	31.22	61.85	26.02	32.22	69.10	35.65
GPT-4o	-	en	29.51	**48.55**	32.35	24.78	52.60	**41.24**	21.85	72.06	32.15
LLaVA-1.5	Vicuna-7B	en	22.65	19.69	23.05	20.21	33.86	38.67	29.26	34.70	22.81
GeoChat	Vicuna-7B	en	22.65	18.96	23.30	20.21	30.33	32.23	-	33.25	22.03
Qwen2-VL	Qwen2-7B	zh	**39.49**	**49.28**	33.26	**37.89**	67.62	24.34	**44.44**	**76.57**	**41.10**
InternVL2	InternLM2.5-7B	zh	34.42	39.12	34.24	37.54	**68.57**	40.09	44.07	76.57	40.58
LLaVA-OneVision	Qwen2-7B	zh	37.48	47.06	31.73	34.72	62.56	29.12	30.37	74.80	38.42
GPT-4o-mini	-	zh	29.51	44.13	29.59	35.41	55.36	**41.86**	21.85	74.71	37.82
InternLM-XComposer-2.5	InternLM2-7B	zh	37.70	43.15	32.62	33.40	62.99	29.47	24.44	69.14	37.26
CogVLM2	Llama3-8B	zh	36.21	45.79	**34.57**	31.03	59.56	26.19	-	70.47	35.84
LLaVA-Next	Llama3-8B	zh	33.08	39.52	31.98	29.34	54.07	21.59	25.56	69.85	33.48
GPT-4o	-	zh	22.28	45.47	31.25	24.93	45.45	27.08	15.19	69.41	30.40
LLaVA-1.5	Vicuna-7B	zh	22.65	19.07	23.01	20.21	33.95	38.67	29.26	37.58	22.86
GeoChat	Vicuna-7B	zh	22.65	19.17	23.05	20.21	24.75	22.79	-	23.74	20.99

Table 3. Detailed image captioning performance on XLRS-Bench.

Method	Language	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L
GeoChat	en	16.74	8.38	4.49	2.45	10.37	16.72
GPT-4o	en	34.69	17.67	8.56	4.04	23.54	20.93
GPT-4o-mini	en	38.29	19.75	9.76	4.29	**23.94**	**21.30**
Qwen2-VL	en	26.74	12.79	5.99	2.53	19.32	19.76
LLaVA-OneVision	en	**41.12**	**20.42**	**9.94**	**4.56**	19.99	21.03
LLaVA-Next	en	27.62	13.45	6.82	3.52	17.78	20.65
LLaVA-1.5	en	35.82	17.62	8.92	4.33	16.49	20.80
CogVLM2	en	30.27	14.46	6.80	3.09	19.37	19.17
InternLM-XComposer-2.5	en	35.17	15.91	7.00	3.02	19.99	17.95
InternVL2	en	25.71	12.44	5.84	2.58	19.55	19.43
GeoChat	zh	6.77	1.49	0.68	0.26	7.84	15.79
GPT-4o	zh	31.08	3.86	1.43	0.43	**26.41**	36.41
GPT-4o-mini	zh	34.13	5.37	2.20	0.58	25.73	**37.11**
Qwen2-VL	zh	21.80	3.50	1.41	0.33	22.92	31.04
LLaVA-OneVision	zh	33.05	5.67	**2.47**	**0.98**	20.24	31.95
LLaVA-Next	zh	13.01	2.10	0.82	0.20	15.12	27.72
LLaVA-1.5	zh	28.56	4.26	1.73	0.00	16.36	29.18
CogVLM2	zh	19.78	2.23	0.79	0.18	22.53	28.33
InternLM-XComposer-2.5	zh	**37.30**	**6.12**	2.39	0.58	20.86	32.97
InternVL2	zh	16.49	3.16	1.39	0.48	22.10	25.76

but this often leads to visual-text misalignment [50] and fails to handle subtleties, contextual nuances, language bias, and diverse question types. These issues are particularly evident in detailed captioning, where long content makes machine translation prone to errors. To build a high-quality Chinese benchmark, we engaged six bilingual experts fluent in Chinese and English to translate and cross-check all long-text content. Finally, 1,400 images, 32,389 QA pairs, 12,619 groundings, and 934 detailed captions are served for the Chinese version of XLRS-Bench, maintaining similar task types, image quality, and difficulty.

4. Experiment

4.1. Experimental Setup

The MLLMs evaluated on XLRS-Bench are grouped into three categories: (a) open-source VLMs, including Qwen2-VL [58], LLava-Onevision [26], LLava-Next [31], LLaVA-1.5 [54], CogVLM2 [17], InternLM-XComposer-2.5 [68] and InternVL-2 [59]; (b) closed-source VLMs, such as GPT-4o [39] and GPT-4o-mini [40]; and (c) the specialized RS model Geochat [22]. For fair comparison, we used a zero-shot setting with uniform prompts for all VLMs. The appendix details the architecture and parameter sizes of each open-source VLM and includes additional results across various settings. Except for GeoChat which was evaluated using its native framework, all other models were evaluated using LMMs-Eval [25, 66]. More results are shown in the Appendix.

4.2. Evaluation Strategy

In the VQA task, we manually created four options for each question: one correct answer and three distractors, derived from text within the image or similar alternatives. This increases the difficulty by requiring models to deeply understand image details. Following MMBench [33] and MME-Realworld [72] methods, we evaluated the accuracy of L-2 capability dimension for the VQA task and reported the average accuracy across L-2 dimensions, with L-3 results available in the appendix. For the L3 sub-task of Overall Land Use Classification, where a question may have multiple correct options, a prediction is considered correct only if it matches the ground truth exactly. For the Grounding task, we used precision, assessing accuracy based on the intersection between predicted and ground truth bounding boxes, with predictions deemed correct if IoU exceeds a threshold. We tested two IoU thresholds: 0.5 and 0.7. For the captioning task, we used standard metrics, including BLEU [43], ROUGE_L [9], and METEOR [4]. We consider the n-gram precision for BLEU with n-values of 1, 2, 3, and 4. All scores in Tables 2, 3, and 4 are reported as percentages (%).

4.3. Main Results

Results on VQA tasks. Table 2 compares various models on the different four L-2 dimensions of perception and reasoning abilities, respectively. Qwen2-VL excels in both English and Chinese proficiency, outperforming both proprietary and most open-source models. Nevertheless, their performance varies across tasks. From the results, we

Table 4. Visual grounding performance on XLRS-Bench and VRSBench [27]. *: VRSBench uses GPT-4V to assess this task.

Benchmark	Method	GPT-4o*	GPT-4o-mini	Qwen2-VL	LLaVA-OneVision	LLaVA-Next	LLaVA-1.5	CogVLM2	InternLM-XComposer-2.5	InternVL2	GeoChat
XLRS-Bench-EN	Acc@0.5	**0.46**	0.09	0.15	0.16	0.18	0.09	0.01	0.02	0.33	0.14
	Acc@0.7	0.05	0.03	0.03	0.00	0.04	0.00	0.00	0.01	**0.12**	0.01
XLRS-Bench-ZH	Acc@0.5	**0.45**	0.21	0.14	0.13	0.07	0.12	0.03	0.06	0.19	0.14
	Acc@0.7	0.03	0.03	0.01	0.01	0.02	0.02	0.00	0.00	**0.06**	0.01
VRSBench	Acc@0.5	5.1	-	-	-	-	-	-	-	-	-
	Acc@0.7	1.1	-	-	-	-	-	-	-	-	-

can draw the following key insights: (1) *GPT-4o's Weakness in Spatiotemporal Reasoning*: GPT-4o underperforms compared to open-source models on complex spatiotemporal reasoning tasks, with accuracy consistently below 25%. These tasks, focused on local counting for change detection, assess MLLMs' ability to capture temporal details across images. GPT-4o likely lacks pretraining on RS scene change detection and sometimes refuses to answer questions citing privacy or harmful content. By contrast, Qwen2-VL performs reliably on these sub-tasks. (2) *Benefits of Higher-Resolution Input Models*: Models like Qwen2-VL, which allow higher-resolution inputs, significantly outperform those using visual encoders like CLIP (*e.g.*, LLaVA1.5). The limit of input image size often necessitates compression, but Qwen2-VL's ability to process richer information at larger image sizes leads to superior performance. (3) *Better Abstract Reasoning in RS*: In high-resolution RS scenarios, models excel at abstract reasoning tasks like anomaly detection, which require minimal local detail and can rely on compressed regional data. Conversely, for perception-heavy tasks requiring fine detail recognition, MLLMs are constrained by their architecture, achieving accuracy rates of only 30%–50%.

Results on image captioning tasks. Our captions have average hl379 words in English and 663 in Chinese, offering richly detailed descriptions with broad applicability. However, evaluations show MLLMs struggle with these lengthy, real-world captions; open-source models like Qwen2-VL and the RS-specific GeoChat perform poorly. Notably, GPT-4o and GPT-4o-mini, even without extensive RS pretraining, excel in long-text generation, significantly outperforming other models. Previous evaluations in RS captioning focused on shorter texts, missing GPT-4o's strengths with longer content. For Chinese caption tasks, GPT-4o still outperforms other models, showcasing its robust understanding and generation abilities as a valuable asset for future research.

Results on grounding tasks. While we anticipated challenges in visual localization during annotation, the poor performance of each MLLM was surprising. To validate the benchmark's effectiveness, we conducted additional human verification: we randomly selected 100 questions from each VQA sub-task (L-3 dimensions) and had two groups answer them concurrently. We also included results from the low-resolution RS visual grounding dataset, VRSBench. Notably, even in 512×512 low-resolution RS scenarios, GPT-4V achieved only 1.1% and 5.1% accuracy; when scaled to resolutions near 8,500×8,500, the accuracy rates of 3.2% and 0.48% are understandable. Some objects in these large images are as small as 5–10 pixels, posing a challenge for current MLLMs.

Limitations and discussions. 1) *Limitations in Spatiotemporal Understanding*: General models (like LLaVA-series and GPT-4o) lack large-scale training tailored to RS spatiotemporal understandings, leading to poor performance on such tasks. For instance, GPT-4o achieves less than 25% accuracy (Table 2). Given the widespread application of these tasks in RS, domain-specific training is essential. Specialized models like GeoChat have yet to support multi-image input tasks, highlingting the need for further research in this area. 2) *Challenges in Ultra-High-Resolution Scenarios*: Current high-resolution MLLMs are limited to 4K images, requiring significant compression for larger-size images, resulting in substantial information loss. This issue is critical for satellite imagery, where the small objects are often measured only 5–10 pixels and shrink to a single pixel after compression, rendering them uninformative. Developing specialized MLLMs for RS super-resolution is anticipated to mitigate this limitation.

5. Conclusion

In this paper, we introduce XLRS-Bench, a comprehensive benchmark for evaluating the perception and reasoning capabilities of multimodal large language models (MLLMs) in ultra-high-resolution remote sensing (RS) scenarios. XLRS-Bench features the largest average image size to date, high-quality human-verified annotations, and 16 sub-tasks across three ability levels, offering a multidimensional evaluation. It supports both English and Chinese, making it the largest manually annotated ultra-high-resolution RS vision-language dataset, surpassing existing benchmarks in data capacity and task diversity. By emphasizing real-world decision-making and spatiotemporal change detection, XLRS-Bench promotes advanced cognitive processes. Experimental results demonstrate that current general-purpose and RS-specific MLLMs still struggle to understand ultra-high-resolution RS imagery, underscoring the need for further improvements.

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